

Materials 1

Chapter 6

Introduction to Phase Diagrams

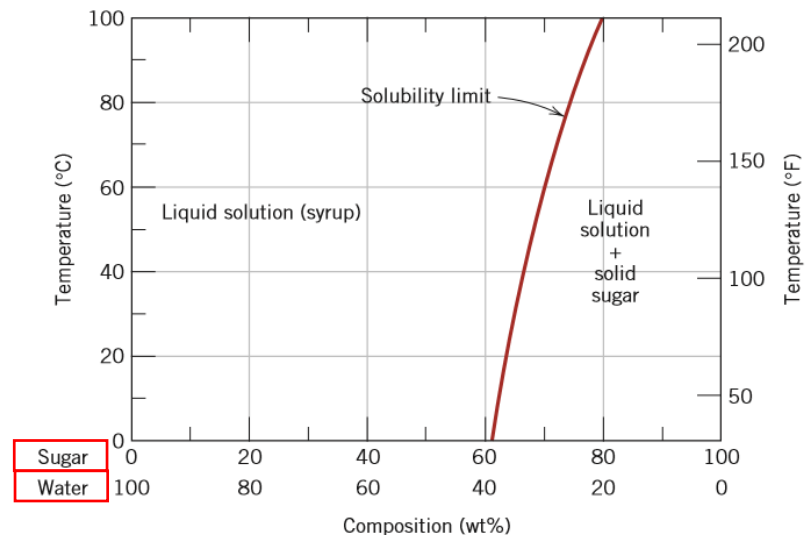
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Definitions

- ❖ **Components** are pure metals and/or compounds of which an alloy is composed.
For example: Cu and Zn in copper–zinc brass .
- ❖ A **system** is defined by the series of possible alloys consisting of the same components, regardless of its alloy composition. For example: iron–carbon system.
- ❖ For many alloy systems and at some specific temperature, there is a maximum concentration of solute atoms that may dissolve in the solvent to form a solid solution; this is called a **solubility limit**.
- ❖ Solubility limit of sugar in water:



Phases

- ❖ A **phase** is defined as a portion of a body of material in which the physical and chemical characteristics are homogeneous.
- ❖ Every **solid**, **liquid**, and **gaseous solution** is considered to be a **phase**. For example: sugar–water solution and solid sugar are 2 different phases.
- ❖ If **more than one phase** is present in a given system, each will have its properties, and a boundary separating the phases will exist across which there will be a discontinuous and abrupt change in the physical and/or chemical characteristics.
- ❖ A single-phase system is called **homogeneous** system.
- ❖ Systems composed of two or more phases are called mixtures or **heterogeneous** systems.

Microstructures

- ❖ The physical properties and mechanical behavior of a material depend on the **microstructure**.
- ❖ The **microstructure** is observed using an optical or electronic microscopic.
- ❖ In metal alloys, the **microstructure** is characterized by the number of phase, their proportions, and the manner in which they are distributed or arranged.
- ❖ For a **two-phase alloy**, one phase may appear light and the other phase dark.
- ❖ When only **a single phase** or solid solution is present, the texture will be uniform, except for grain boundaries that may be revealed

Phase Equilibria

- ❖ A system is at **equilibrium** if its free energy is at a minimum under some specified combination of temperature, pressure, and composition.
- ❖ **Phase equilibrium** refers to the equilibrium of systems in which more than one phase may exist.
Example of phase equilibrium: sugar-syrup for a liquid-solid system
- ❖ In many metallurgical and materials systems, **phase equilibrium** involves just solid phases.
- ❖ **Non-equilibrium or metastable state** is defined when a system does not completely achieve a state of equilibrium since the rate of approach to equilibrium is extremely slow.
- ❖ **Metastable structures** are of more practical significance than equilibrium ones.
For example: some steel and aluminum alloys rely for their strength on the development of metastable microstructures during heat treatments.

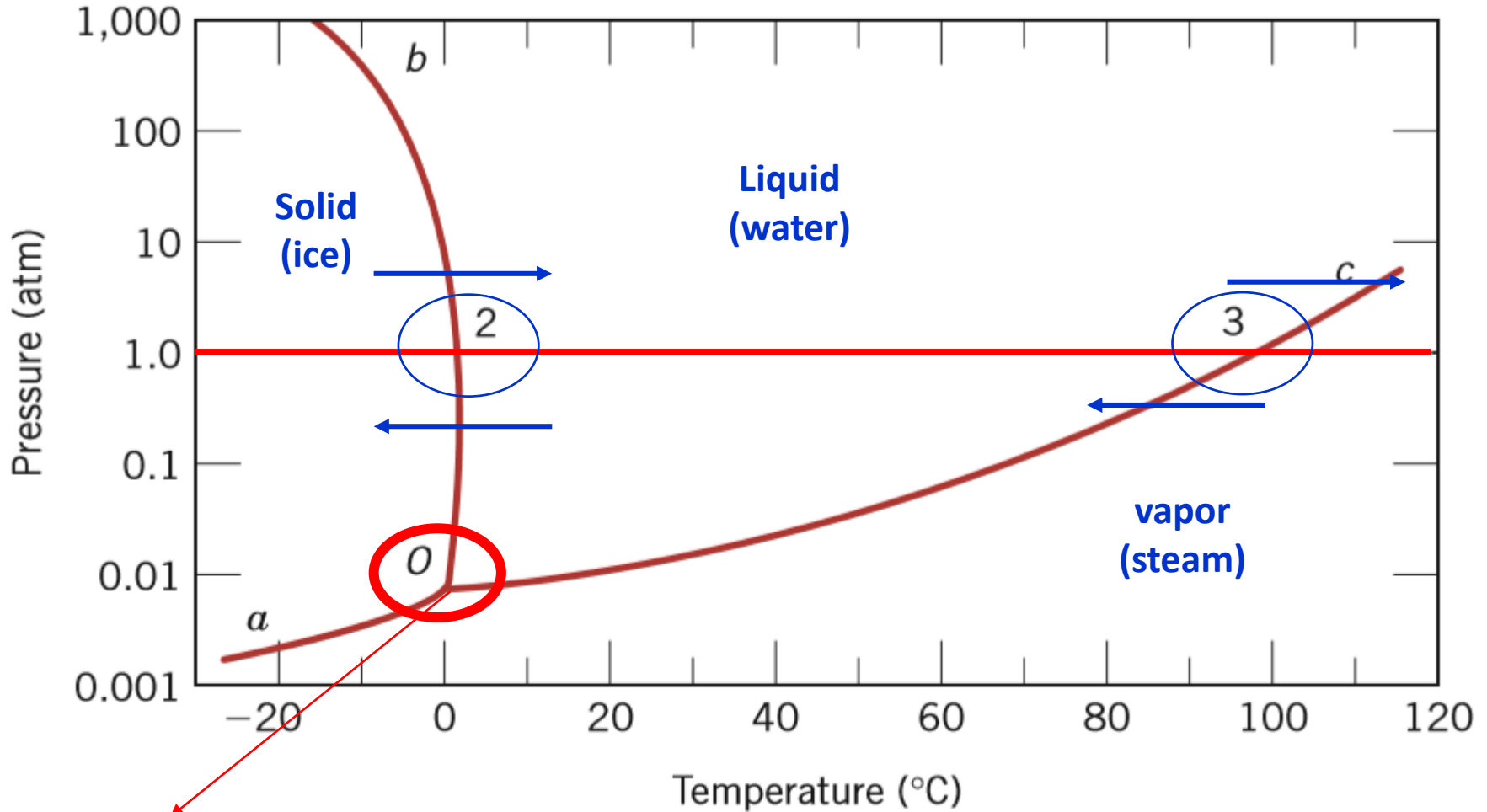
Unary Phase Diagrams

- ❖ **Phase diagram** or **equilibrium diagram** provides information about the control of the phase structure of a particular system.
- ❖ The **phase diagram** is constructed when the 3 externally controllable parameters, **temperature**, **pressure**, and **composition** are plotted against each other.
- ❖ The **unary phase diagram**, also called the **one-component** or the pressure-temperature (P-T) diagram, is a type of phase diagrams in which the composition is held constant and the pressure and temperature are the variables.

Example of unary phase equilibrium: Phase diagram for a pure substance.

Unary Phase Diagrams

Phase Diagram for pure H₂O



The 3 phases, solid, liquid and vapor, are in equilibrium (triple point).

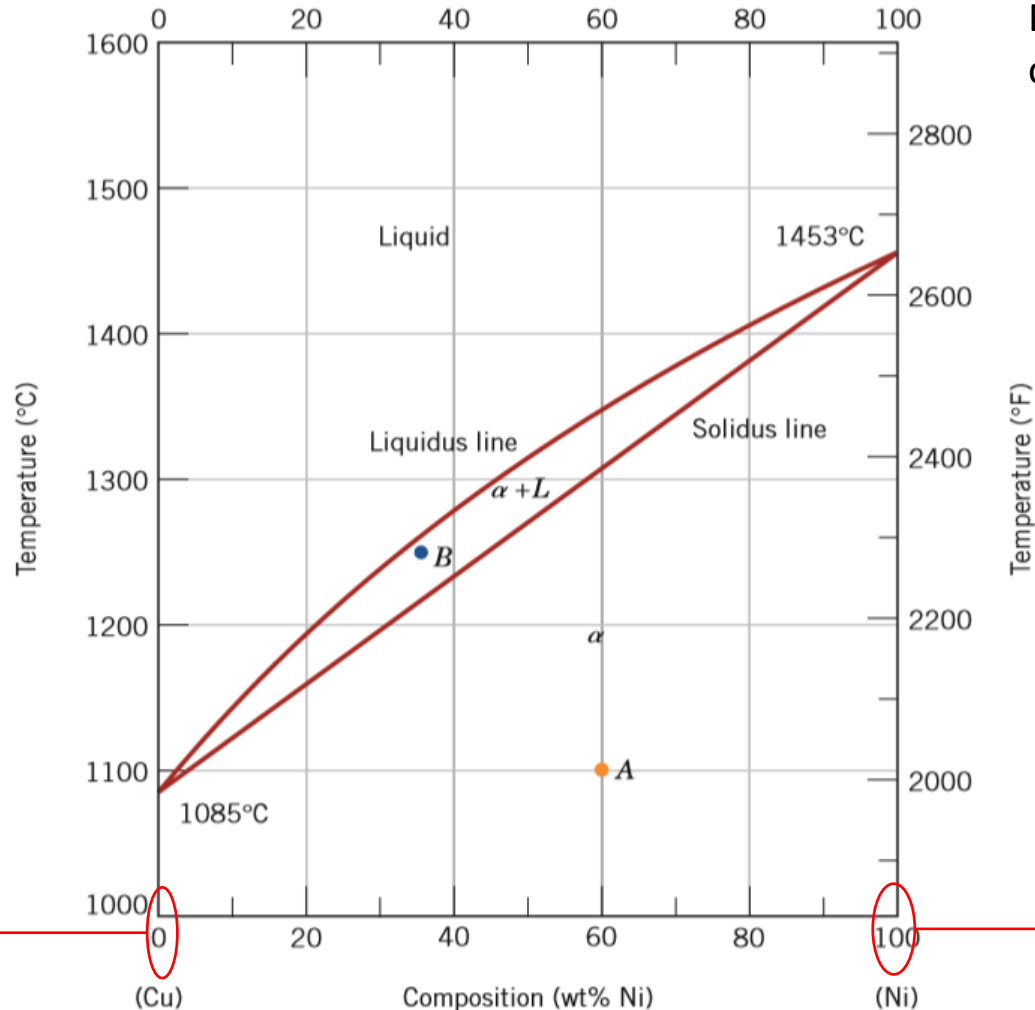
Binary Phase Diagrams

- ❖ A **binary phase diagram** is a phase diagram in which the temperature and composition parameters are variable; however, the pressure is constant (1 atm).
- ❖ A **binary phase diagram** provides information about binary alloys which are alloys that contain two components.
- ❖ **Binary phase diagrams** represent the relationship between temperature, the compositions and quantities of phases at equilibrium, which influence the microstructure of an alloy.
- ❖ Many **microstructures** develop from phase transformations that occur when the temperature is altered (usually upon **cooling**).
 - ➡ This may involve the transition from one phase to another, or the appearance or disappearance of a phase.
- ❖ **Binary phase diagram** predicts phase transformations and the resulting microstructures, which may have equilibrium or non-equilibrium character.

Binary Phase Diagrams

Copper-Nickel Phase Diagram

Composition (at% Ni)



3 phase regions or fields:

- an alpha (α) field
- a liquid (L) field
- a two-phase alpha and liquid field ($\alpha + L$)

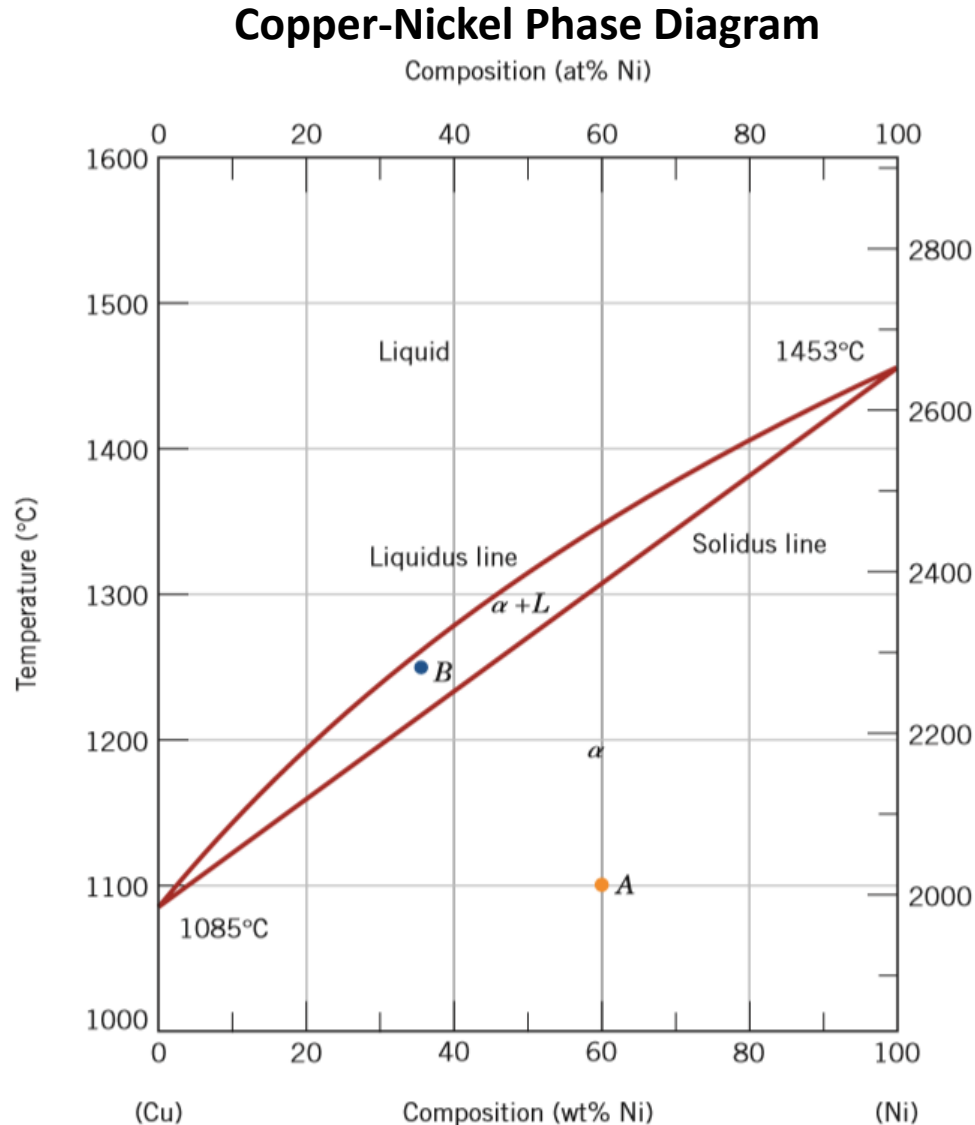
Each region is defined by one or more phase(s):

- liquid (L) phase
- alpha (α) phase

0 wt% of Ni and
100 wt% Cu

0 wt% of Cu and
100 wt% Ni

Binary Phase Diagrams



Complete solubility of Cu-Ni alloy in the solid solution (α) below 1085°C:

- Both Ni and Cu have the same crystal structure (FCC).
- Both have almost identical radii.
- Both have similar electronegativity and valences.



Isomorphous:
complete liquid and solid solubility of the two components.

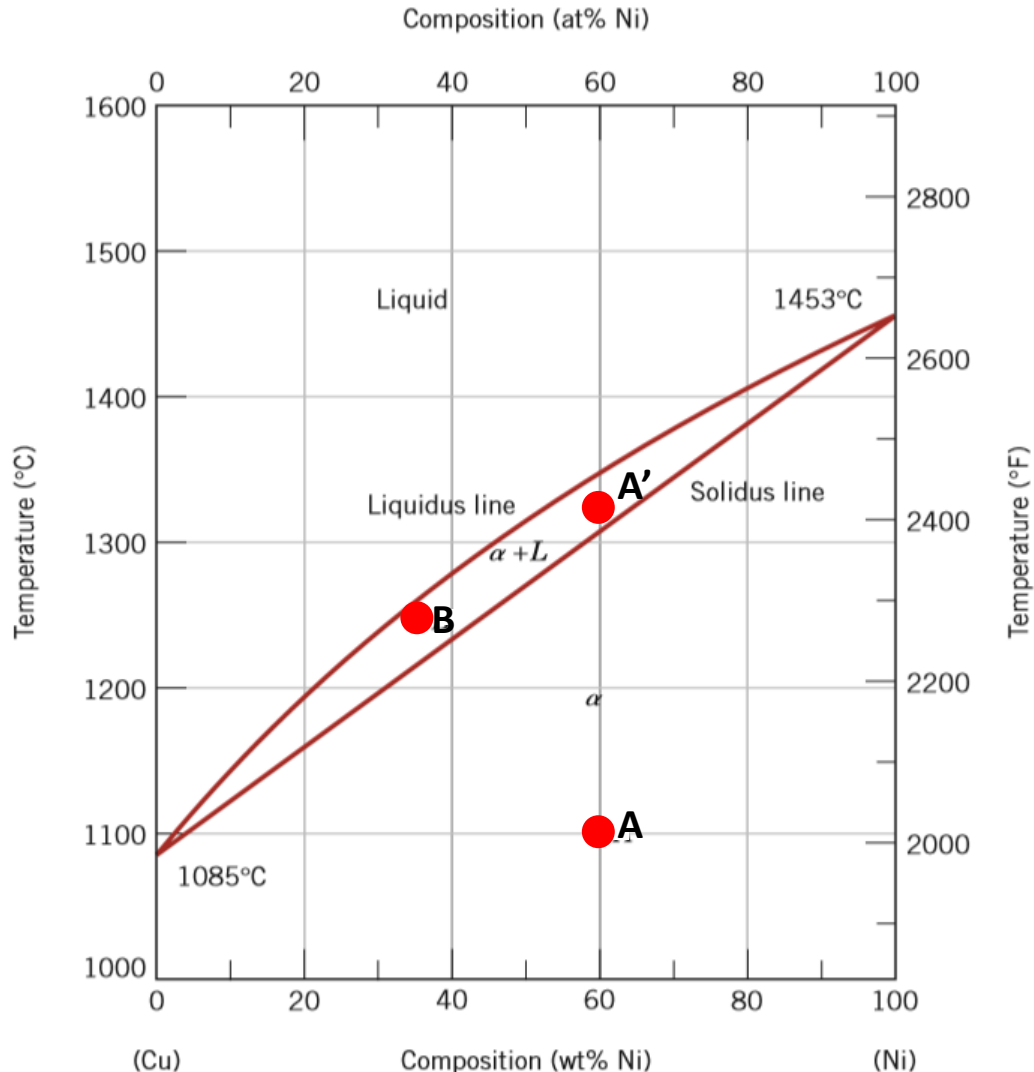
Interpretation Phase Diagrams

From a binary phase diagram, we can determine:

- The phases that are present.
- The composition of phases.
- The percentage and fraction of the phases.

Interpretation Phase Diagrams

Copper-Nickel Phase Diagram



Point A at 1110°C:

- ✓ Cu-Ni alloy composition: 40 wt% of Cu and 60 wt% Ni
- ✓ Consists of a single α phase.

Point A' at 1320°C :

- ✓ Cu-Ni alloy composition: 40 wt% of Cu and 60 wt% Ni
- ✓ Consists of both α and liquid phases.

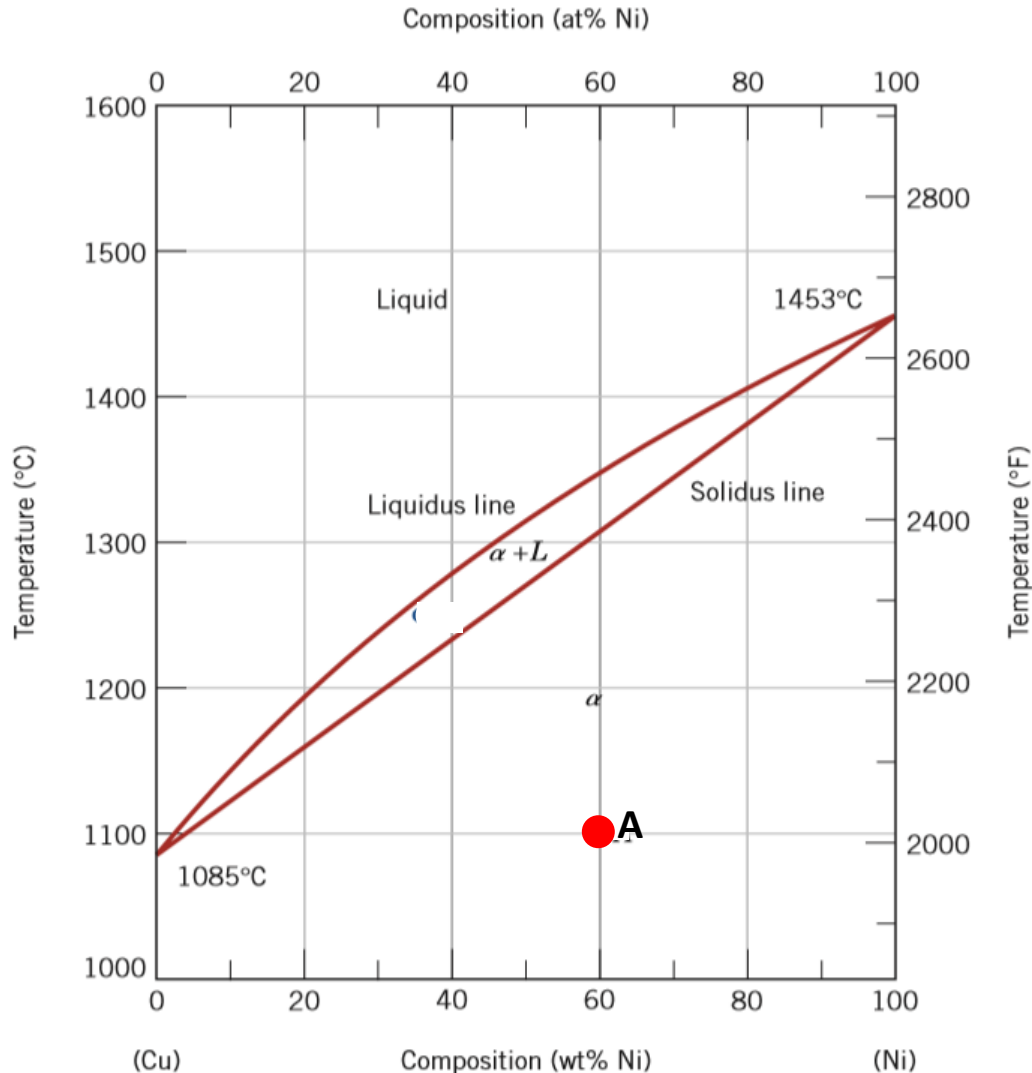
➡ Same composition for 2 different phase regions.

Point B at 1250°C:

- ✓ Cu-Ni alloy composition: 65 wt% of Cu and 35 wt% Ni
- ✓ Consists of both α and liquid phases.

Interpretation Phase Diagrams

Copper-Nickel Phase Diagram



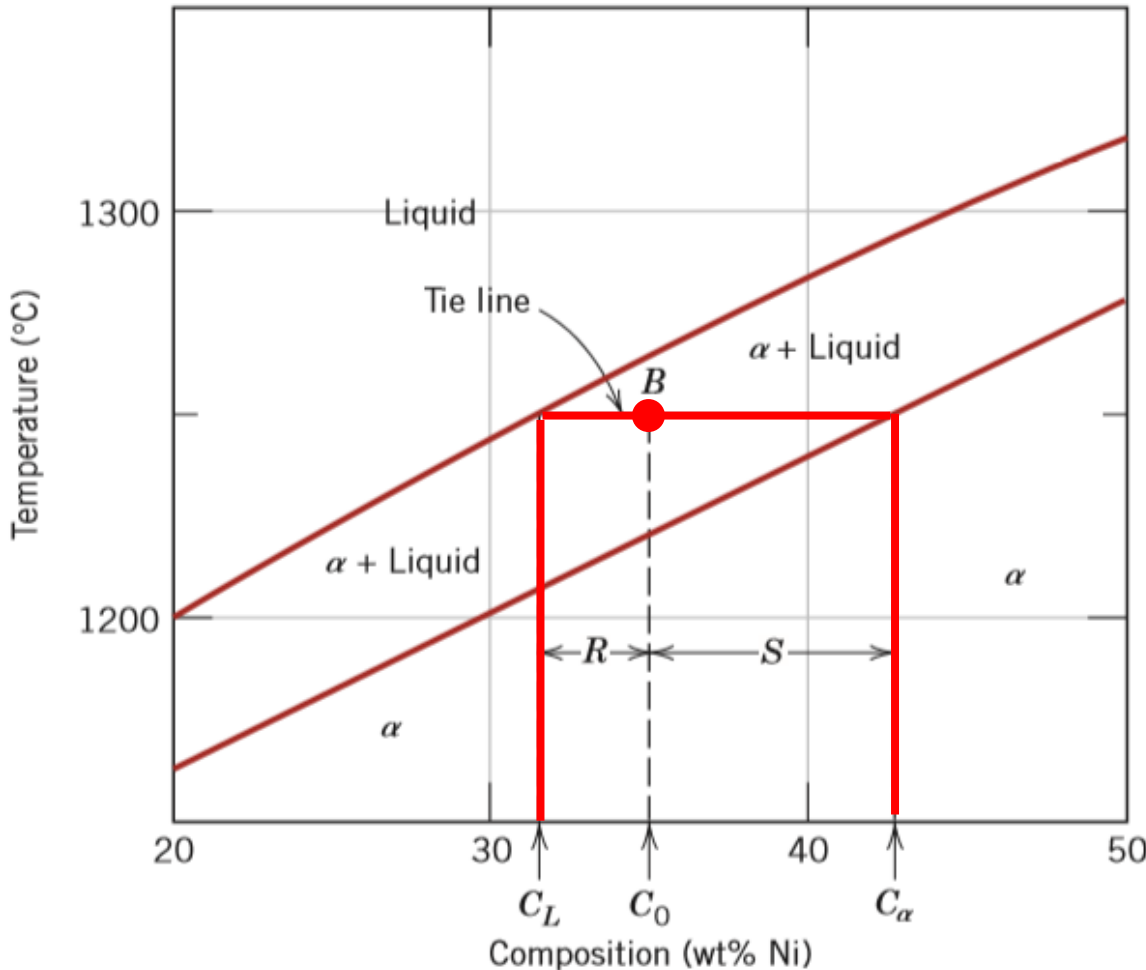
For a **single phase region**, the **phase composition** is the same as the **overall composition of the alloy**.

Point A at 1110°C:

- ✓ single α phase
- ✓ **Phase composition:** 40 wt% of Cu and 60 wt% Ni

Interpretation Phase Diagrams

Copper-Nickel Phase Diagram



For a **two phase region**, point B at 1250°C :
 $C_0 = 35 \text{ wt\% Ni}$ and $65 \text{ wt\% Cu}$

Steps to compute the **phases composition**:

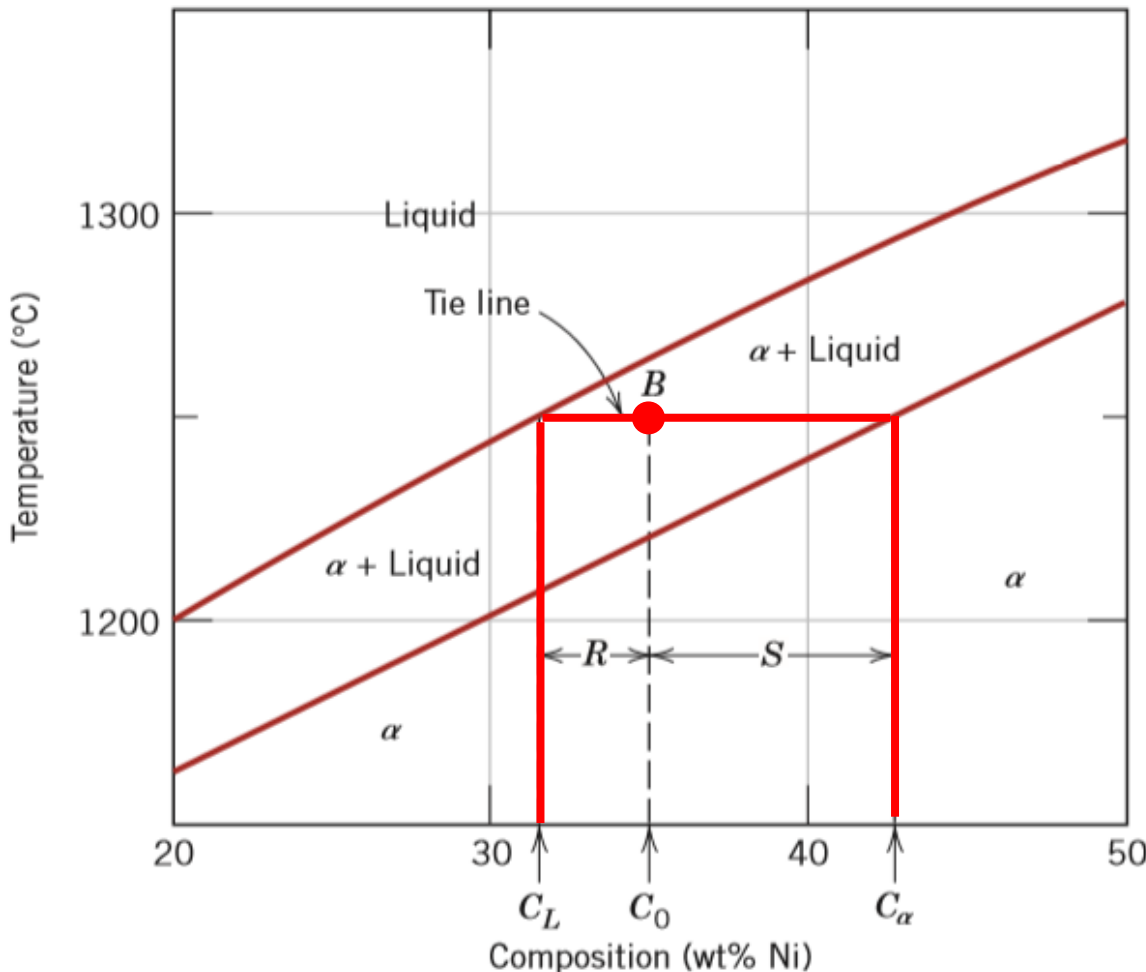
- 1) A straight line is constructed across the two-phase region at the temperature of the alloy.
- 2) At the intersection of the line with each of the phase boundaries, a perpendicular line at the intersection is drawn and the composition of each is determined.

$$C_L = 31.5 \text{ wt\% Ni and } 68.5 \text{ wt\% Cu}$$

$$C_\alpha = 42.5 \text{ wt\% Ni and } 57.5 \text{ wt\% Cu}$$

Interpretation Phase Diagrams

Copper-Nickel Phase Diagram



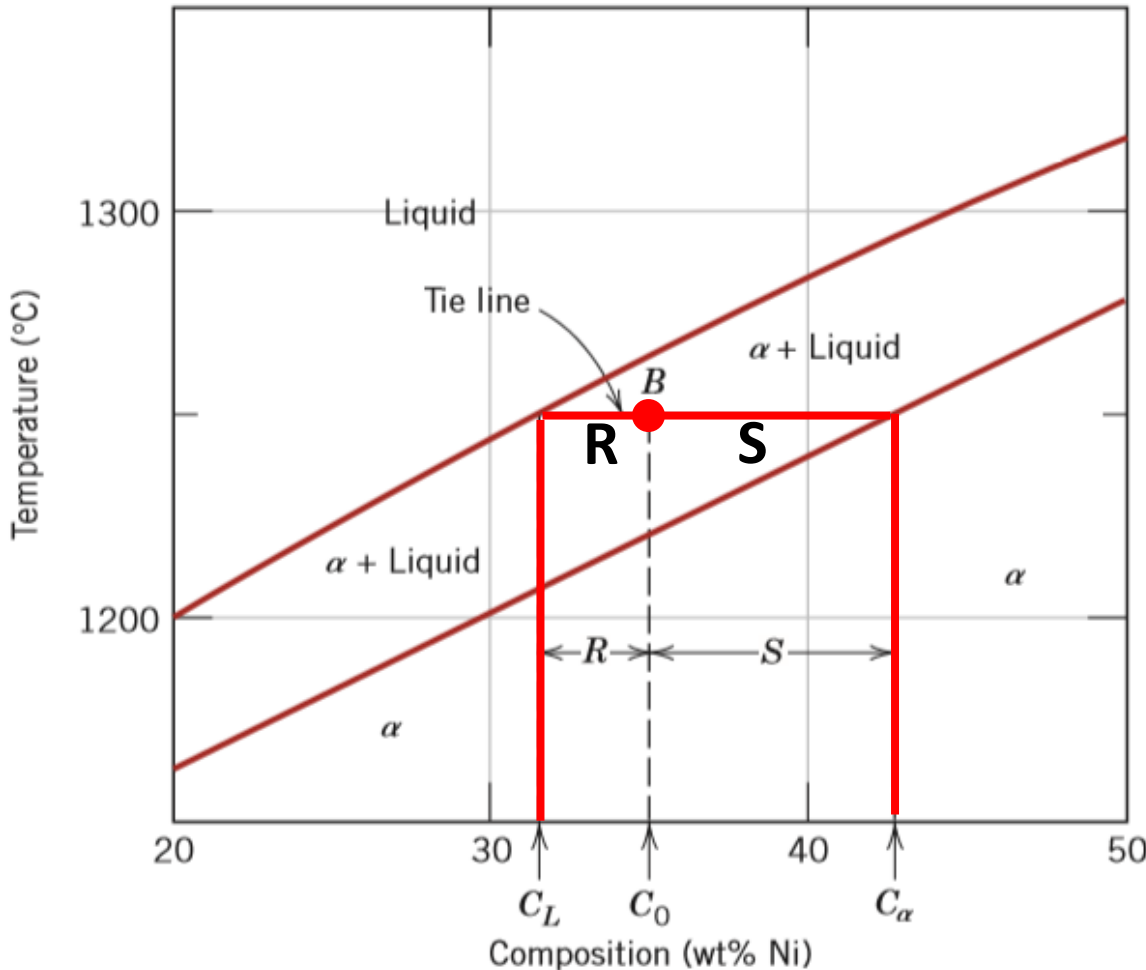
- 3) The fraction of one phase is computed by taking the length of straight line from the overall alloy composition to the phase boundary for the other phase, and dividing by the total length of the straight line.

Lever Rule

- 4) The mass (or weight) of each phase is computed from the product of each phase fraction and the total alloy mass (weight).

Interpretation Phase Diagrams

Copper-Nickel Phase Diagram



Lever Rule

$$W_L = \frac{S}{R + S}$$

Or by subtracting compositions:

$$W_L = \frac{C_\alpha - C_0}{C_\alpha - C_L}$$

$C_0 = 35 \text{ wt\% Ni and } 65 \text{ wt\% Cu}$

$C_L = 31.5 \text{ wt\% Ni and } 68.5 \text{ wt\% Cu}$

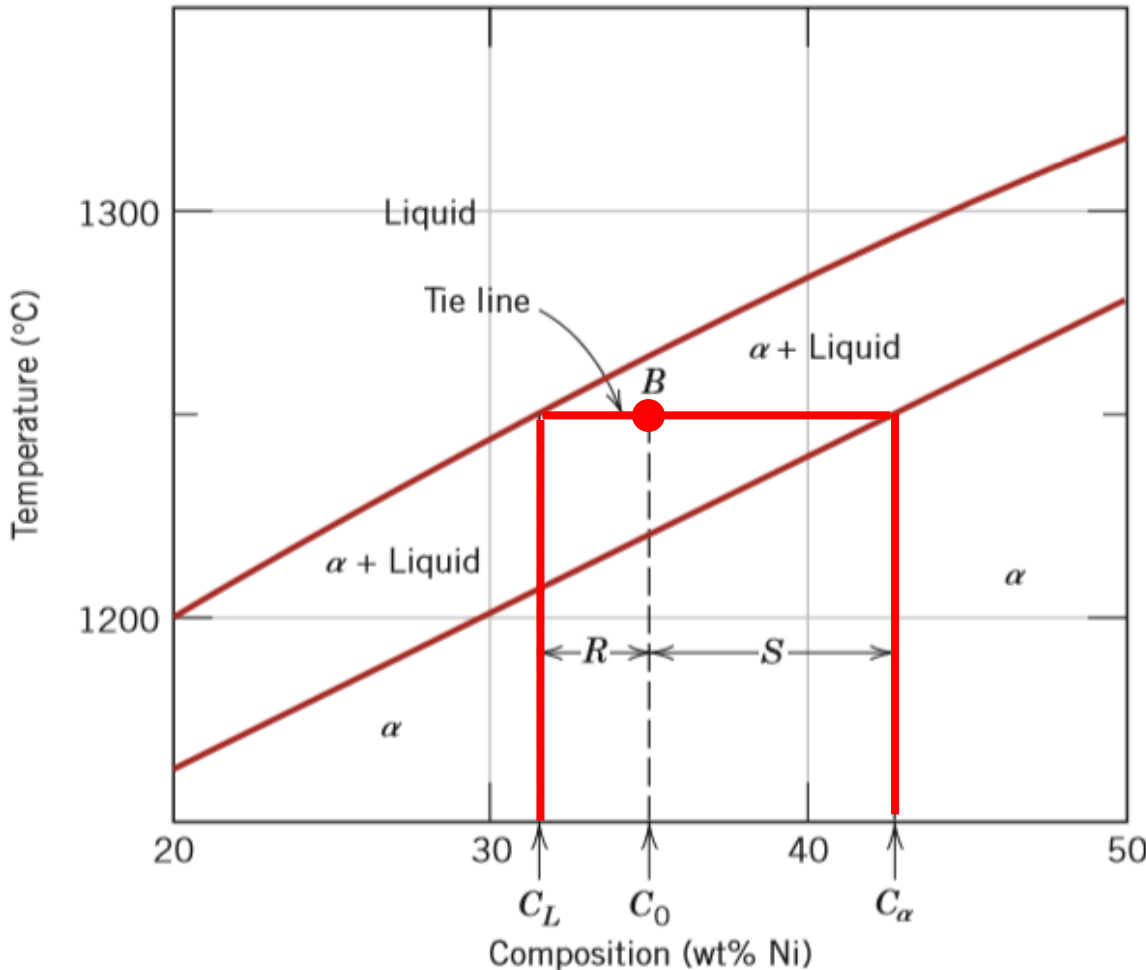
$C_\alpha = 42.5 \text{ wt\% Ni and } 57.5 \text{ wt\% Cu}$

Then, the mass fraction of liquid phase:

$$W_L = 0.68$$

Interpretation Phase Diagrams

Copper-Nickel Phase Diagram



Lever Rule

$$W_\alpha = \frac{R}{R + S}$$

Or by subtracting compositions:

$$W_\alpha = \frac{C_0 - C_L}{C_\alpha - C_L}$$

$C_0 = 35 \text{ wt\% Ni and } 65 \text{ wt\% Cu}$

$C_L = 31.5 \text{ wt\% Ni and } 68.5 \text{ wt\% Cu}$

$C_\alpha = 42.5 \text{ wt\% Ni and } 57.5 \text{ wt\% Cu}$

Then, the mass fraction of α phase:

$$W_\alpha = 0.32$$

Interpretation Phase Diagrams

- ❖ For multiphase alloys, **phase volume fractions** are preferred rather than **mass fractions** because they may be determined from the examination of the microstructure.
- ❖ Moreover, the properties of a multiphase alloy may be estimated on the basis of volume fractions.
- ❖ Therefore, it is more convenient to specify the **phase composition** in terms of **volume fraction** rather than **mass fraction** for multiphase alloys.

Interpretation Phase Diagrams

Phase Volume Fraction

❖ Consider an alloy consisting of α and β phases.

❖ The volume fraction of the α phase, V_α :

$$V_\alpha = \frac{v_\alpha}{v_\alpha + v_\beta}$$

❖ The volume fraction of the β phase, V_β :

$$V_\beta = \frac{v_\beta}{v_\alpha + v_\beta}$$

Where:

v_α : volume of α phase in the alloy

v_β : volume of β phase in the alloy

Interpretation Phase Diagrams

Phase Volume Fraction

Conversion of mass fractions of α and β phases to volume fractions:

➤ The volume fraction of the α phase, V_α :

$$V_\alpha = \frac{\frac{W_\alpha}{\rho_\alpha}}{\frac{W_\alpha}{\rho_\alpha} + \frac{W_\beta}{\rho_\beta}}$$

➤ The volume fraction of the β phase, V_β :

$$V_\beta = \frac{\frac{W_\beta}{\rho_\beta}}{\frac{W_\alpha}{\rho_\alpha} + \frac{W_\beta}{\rho_\beta}}$$

Where:

ρ_α : density of α phase in the alloy

ρ_β : density of β phase in the alloy

Interpretation Phase Diagrams

Phase Mass Fraction

Conversion of volume fractions of α and β phases to mass fractions:

- The mass fraction of the α phase, W_α :

$$W_\alpha = \frac{\rho_\alpha V_\alpha}{\rho_\alpha V_\alpha + \rho_\beta V_\beta}$$

- The mass fraction of the β phase, W_β :

$$W_\beta = \frac{\rho_\beta V_\beta}{\rho_\alpha V_\alpha + \rho_\beta V_\beta}$$

Thank you for your attention!